Prueba Objetiva 1

1) $f(x) = \dfrac{x^2}{1 - x^2}$

Asint. V.

$1 - x^2 = 0$
$-x^2 = -1$
$x = \pm\sqrt{1}$
$x = \pm 1$

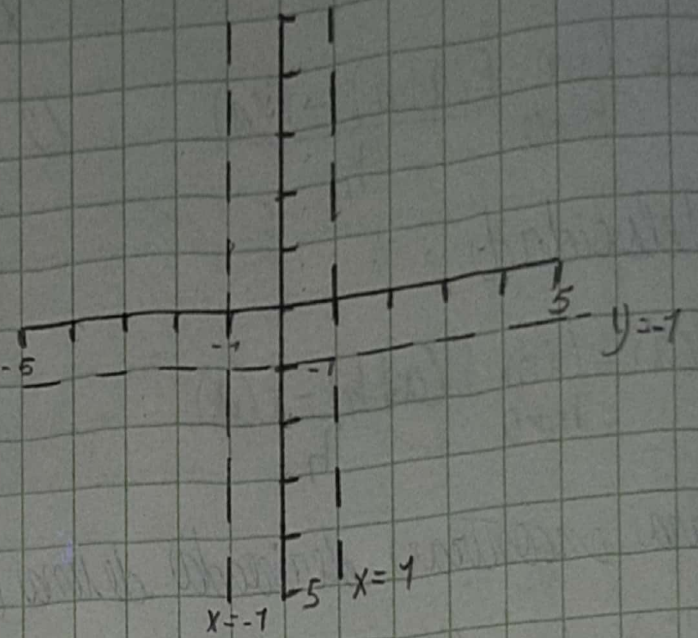
Ecua.

$x = -1$
$x = 1$

Asint. Hz.

$\dfrac{1}{-1} = -1$

Ecua.
$y = -1$



2) $\lim\limits_{v \to 3} \dfrac{x^2 - 9}{sen(x-3)}$ $v = x$

$\lim\limits_{x \to 3} \dfrac{x^2 - 9}{sen(x-3)} = \dfrac{3^2 - 9}{sen(3-3)} = \dfrac{0}{0}$ forma indeterminada

$\lim\limits_{x \to 3} \dfrac{x^2 - 9}{sen(x-3)} = \dfrac{(x-3)(x+3)}{cos(x-3)} = \dfrac{(3-3)(3+3)}{cos(3-3)} = \dfrac{6}{1} = 6$

3) $\lim\limits_{x \to 0} \dfrac{\sqrt[3]{x+64} - 4}{x} = \dfrac{4-4}{0} = \dfrac{0}{0}$ forma indeterminada

$\lim\limits_{x \to 0} \dfrac{\sqrt[3]{x+64} - 4}{x} \cdot \dfrac{\sqrt[3]{(x+64)^2} + 4\sqrt[3]{x+64} + 16}{\sqrt[3]{(x+64)^2} + 4\sqrt[3]{x+64} + 16} = \dfrac{x + 64 - 64}{x(\sqrt[3]{(x+64)^2} + 4\sqrt[3]{x+64} + 16)}$

$\lim\limits_{x \to 0} \dfrac{\cancel{x}}{\cancel{x}(\sqrt[3]{(x+64)^2} + 4\sqrt[3]{x+64} + 16} = \dfrac{1}{\sqrt[3]{(x+64)^2} + 4\sqrt[3]{x+64+16}} = \dfrac{1}{\sqrt[3]{(0+64)^2} + 4\sqrt[3]{0+64} + 16}$

$\dfrac{1}{48} \approx 0.02$

4) $h(t) = 4t - 5t^2$

$$V(a) = \lim_{h\to 0} \frac{4(a+h) - 5(a+h)^2 - (4a - 5a^2)}{h} =$$

$$\lim_{h\to 0} \frac{\cancel{4a} + 4h - 5(a^2 + 2ah + h^2) - \cancel{4a} + 5a^2}{h} =$$

$$\lim_{h\to 0} \frac{4h - \cancel{5a^2} - 10ah - 5h^2 + \cancel{5a^2}}{h} = \frac{\cancel{h}(4 - 10a - 5h)}{\cancel{h}} = \lim_{h\to 0} 4 - 10a - 5h$$

$$V(a) = 4 - 10a$$

¿Cuando caerá?:

$0 = 4t - 5t^2$ $\quad 0 = t \quad 0 = 4 - 5t$

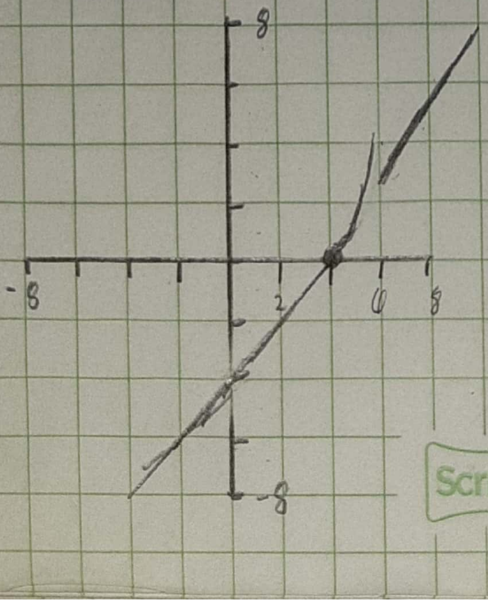
$0 = t(4 - 5t)$ $\quad -4 = -5t$

$4/5 = t$

Velocidad de cuando cae al suelo:

$$V\left(\frac{4}{5}\right) = 4 - 10\left(\frac{4}{5}\right) = -4 \qquad \text{Rapidez} = 4$$

5)
$$f(x) = \begin{cases} x - 4 & x < 4 \\ (x-4)^2 & \text{si } 4 \leq x < 6 \\ x - 3 & x \geq 6 \end{cases}$$



Es discontinua en:

$x = 6$, de salto

Continua

$x = 4$

4) $f(x) = \dfrac{1}{x+5}$ $\qquad x = -4$

$f(-4) = \dfrac{1}{-4+5} = \dfrac{1}{1} = 1$ $\qquad \underline{P(-4, 1)}$

$f'(a) = \lim_{x \to -4} \dfrac{\frac{1}{x+5} - 1}{x+4} = \dfrac{0}{0}$ forma indeterminado

$$\lim_{x \to -4} \dfrac{\frac{1-(x+5)}{x+5}}{x+4} = \dfrac{\frac{1-x-5}{x+5}}{x+4} = \dfrac{\frac{-4-x}{x+5}}{x+4} = -\dfrac{1}{x+5}$$

$$\lim_{x \to -4} -\dfrac{1}{-4+5} = \underline{-1 = m}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -1(x - (-4))$$

$$y = \underline{-x - 3}$$